

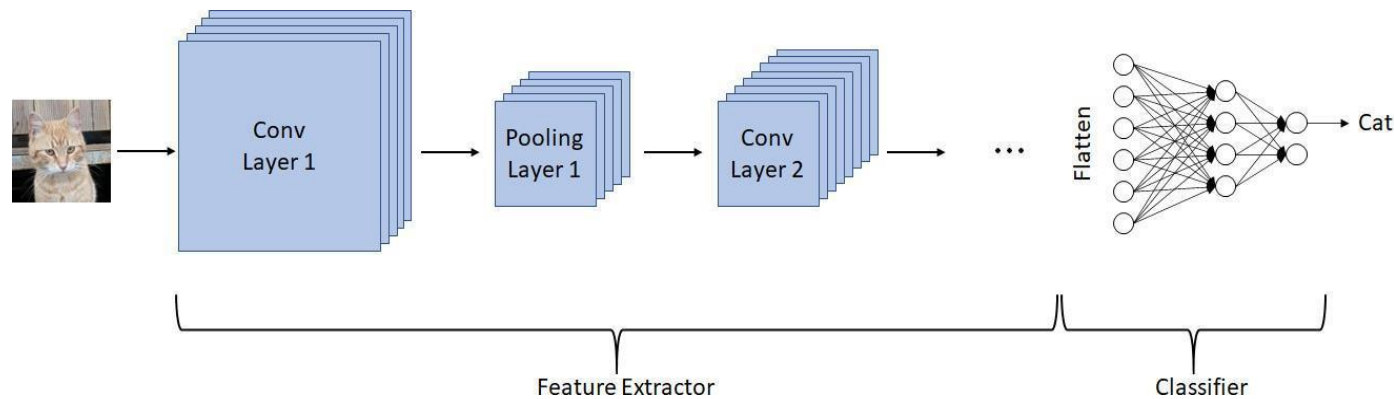
COMP4616 – Fundamentals of Machine Learning

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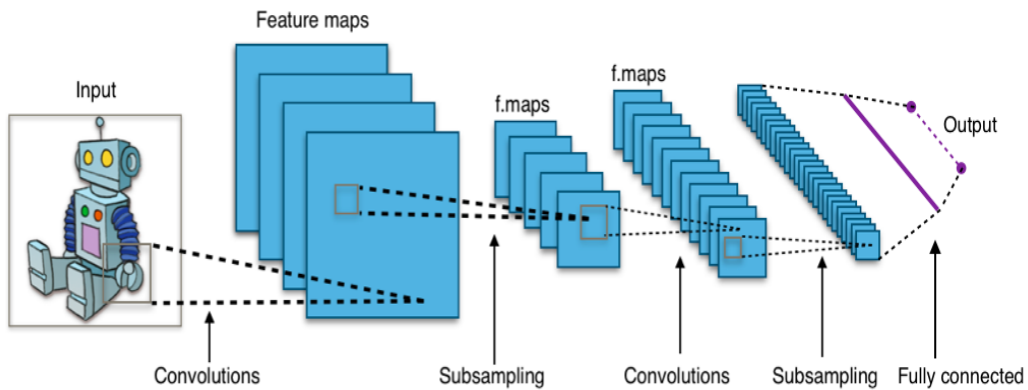
SPRING 2026

TYPES OF AN - PART 1: CONVOLUTIONAL NEURAL NETWORKS (CNN) & RECURRENT NEURAL NETWORKS (RNN)

Convolutional Neural Networks



What is a Convolutional Neural Network?



- CNN is a multilayer feedforward ANN modeled on the structure of the animal visual cortex
- Key layers
 - **Convolutional layers:** learn spatial features
 - **Pooling layers:** downsample & retain important features
 - **Fully connected layers:** final classification

Historical Milestones

- 1959** Hubel & Wiesel: visual cortex "receptive fields"
- 1980** Neocognitron: S-cells (features) & C-cells (invariance)
- 1989** LeCun et al.: first successful backprop CNN for zip codes
- 1998** LeNet: modern CNN architecture (sigmoid, backprop)

Convolutional Layer

Automatic Feature Learning

Filters/kernels slide over the input and perform elementwise multiplication followed by summation, producing a feature map.

Local Connectivity

Each filter focuses on a small local region, capturing spatially invariant patterns (e.g., edges regardless of position).

Weight Sharing

The same filter is applied across all positions of the input, drastically reducing parameters and preventing overfitting.

Keras Example

```
model.add(Conv2D(32, (3,3), activation='relu',  
input_shape=(28,28,1)))
```

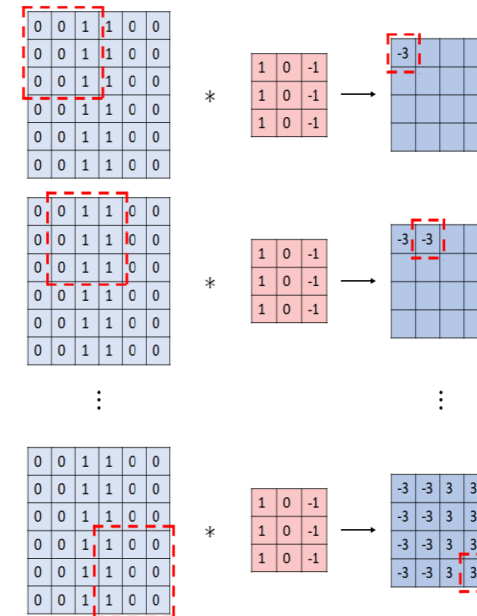
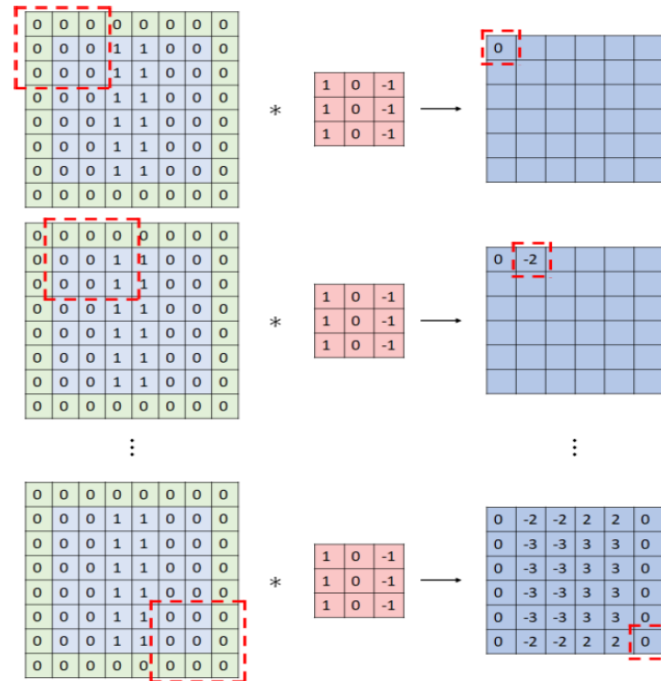


Figure 2.5 Visualization of convolution operation on with a 5×5 image and a 3×3 filter with a stride of 1 pixel.

Convolution Operation & Padding



Output Size Formula

$$\text{Output size} = \frac{W - K + 2P}{S} + 1$$

W: Input size

K: Filter (kernel) size

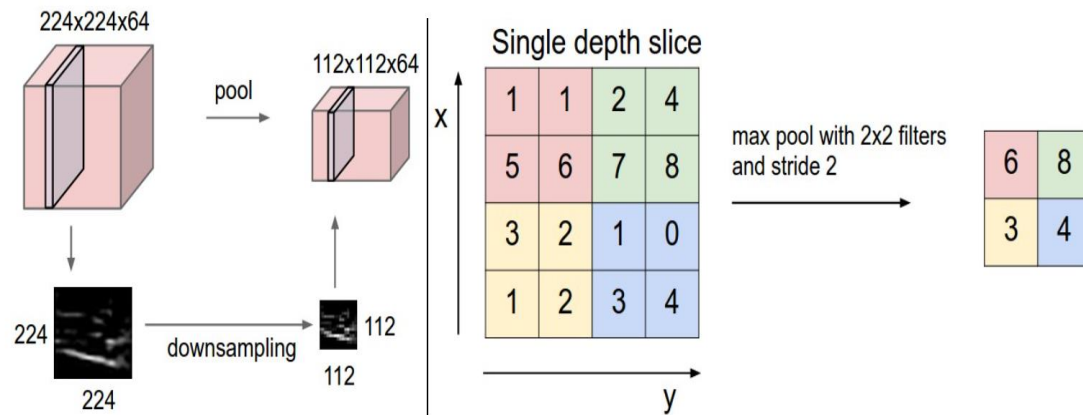
S: Stride (step size)

P: Padding (extra pixels)

Padding preserves spatial dimensions and allows the filter to reach border pixels.

Figure 2.6 Visualization of convolution operation on with a 5×5 image and a 3×3 filter with a stride of 1 pixel and zero padding of 1 pixel as per Equation 2.2. Green background is padding while the image data is shown with light blue background.

Convolution Operation & Padding



Pooling layer downsamples the volume spatially, independently in each depth slice of the input volume. **Left:** In this example, the input volume of size $[224 \times 224 \times 64]$ is pooled with filter size 2, stride 2 into output volume of size $[112 \times 112 \times 64]$. Notice that the volume depth is preserved. **Right:** The most common downsampling operation is max, giving rise to **max pooling**, here shown with a stride of 2. That is, each max is taken over 4 numbers (little 2×2 square).

Purpose of Pooling

- Downsamples feature maps, reducing spatial dimensions
- Retains dominant features while cutting computation
- Provides a degree of translation invariance
- Most common: Max Pooling (takes maximum in each region)
- Average Pooling also used (less common)

Max Pooling

Takes the maximum value from each receptive field. Preserves the most prominent features.

Average Pooling

Takes the average value from each receptive field. Smoother representation, less used.

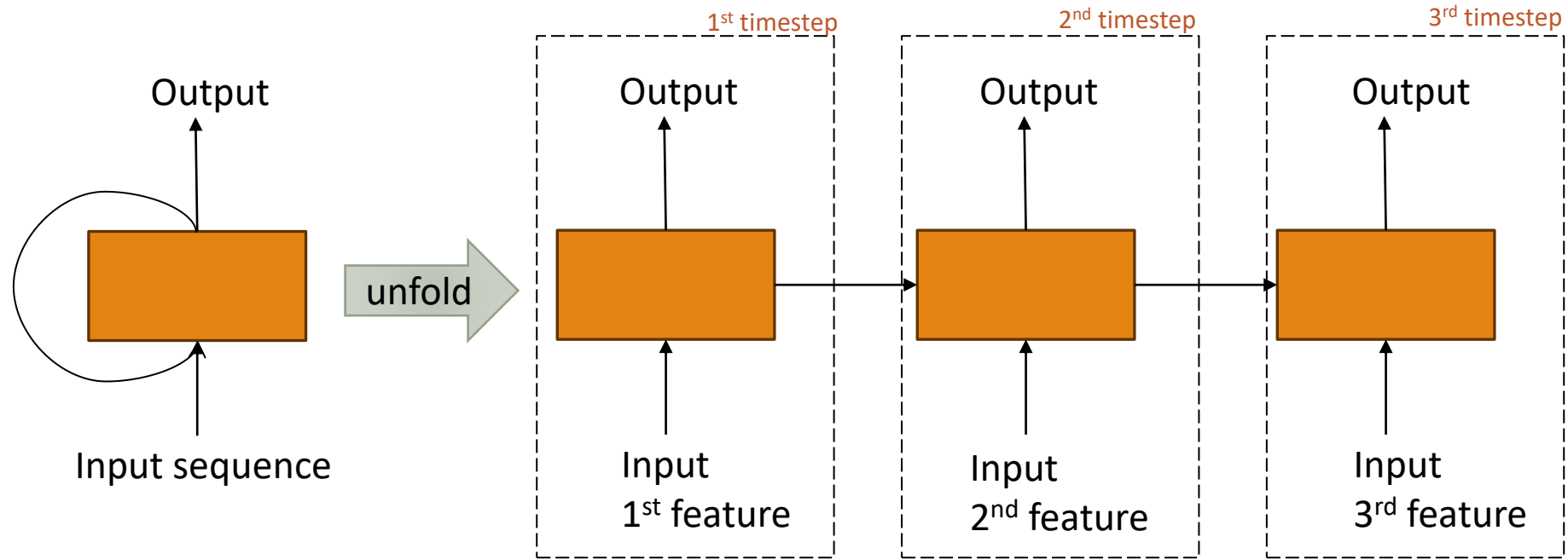
CNN: Advantages & Disadvantages

Advantages

- ✓ Automated feature extraction (no manual engineering)
- ✓ High accuracy on image recognition tasks
- ✓ Transfer learning: adapt pretrained models with minimal retraining
- ✓ Spatially invariant feature detection via weight sharing

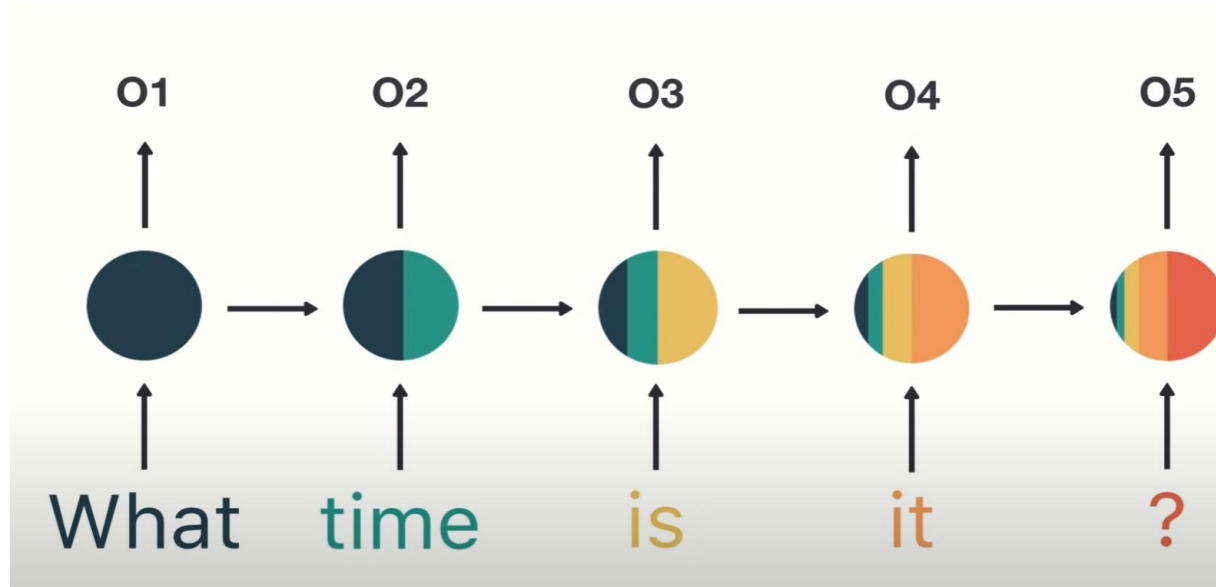
Disadvantages

- ✗ High computational & memory requirements
- ✗ Needs large labeled datasets. Poor performance on small data
- ✗ Vulnerable to adversarial examples
- ✗ Difficult to interpret internal representations



Recurrent Neural Networks

What is a Recurrent Neural Network?



RNNs are neural networks designed for **sequential data** where the order of inputs matters.

Unlike feedforward networks:

- Each hidden unit has connections back to itself from the previous time step
- This creates an internal **memory** of past inputs
- The network can capture temporal dependencies

What is Sequential Data?

Time Series: Stock prices, weather

Text Data: Natural language

Speech: Sound waves, phonemes

Video: Sequences of frames

What is a Recurrent Neural Network?

SimpleRNN class

```
keras.layers.SimpleRNN(  
    units,  
    activation="tanh",  
    use_bias=True,  
    kernel_initializer="glorot_uniform",  
    recurrent_initializer="orthogonal",  
    bias_initializer="zeros",  
    kernel_regularizer=None,  
    recurrent_regularizer=None,  
    bias_regularizer=None,  
    activity_regularizer=None,  
    kernel_constraint=None,  
    recurrent_constraint=None,  
    bias_constraint=None,  
    dropout=0.0,  
    recurrent_dropout=0.0,  
    return_sequences=False,  
    return_state=False,  
    go_backwards=False,  
    stateful=False,  
    unroll=False,  
    seed=None,  
    **kwargs  
)
```

At each timestep t

Receive input x_t

The network receives the current element in the sequence.

Update hidden state h_t

$$h_t = f(W_{hx}x_t + W_{hh}h_{t-1} + b_h)$$

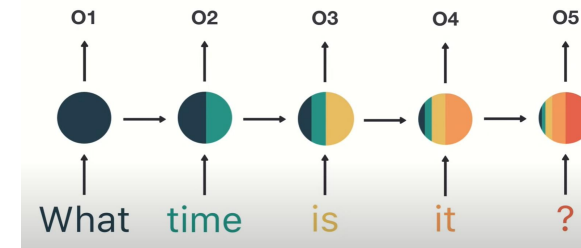
The hidden state combines current input with memory from the previous step.
 f is the activation function. Usually tanh.

Produce output o_t

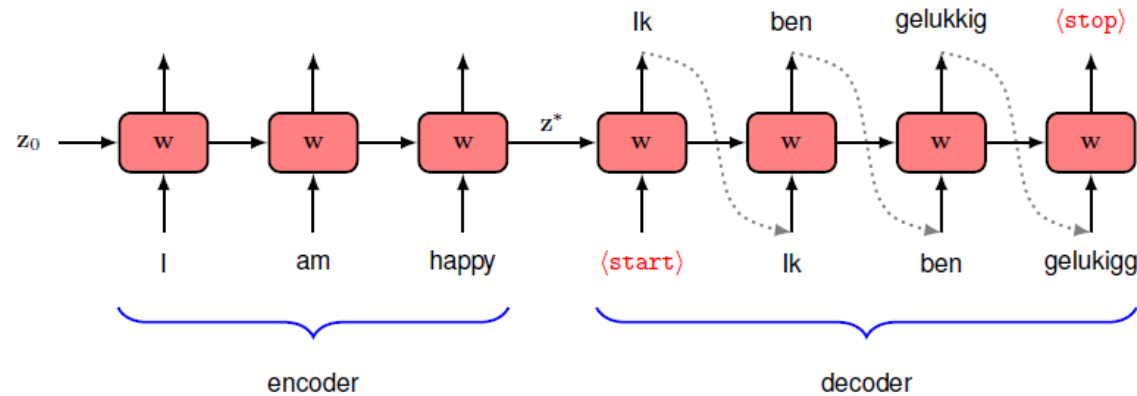
An output may be generated at each step (many to many) or only at the final step (many to one).

BPTT Training

Backpropagation Through Time unrolls the network across timesteps and applies standard gradient descent.



Example

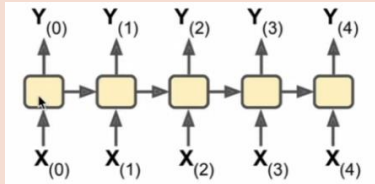


- Assume that we want to convert English sentences to Dutch.
- Feed in the complete English sentence followed by a special input token, which we denote by $\langle \text{start} \rangle$, to trigger the start of translation.
- During training the network learns to associate $\langle \text{start} \rangle$ with the beginning of the output sentence.
- We also take each successively generated word and feed it into the input at the next time step

What Can RNNs Do?

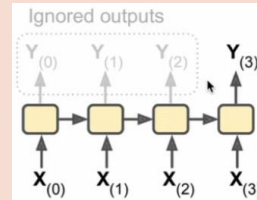
Many $\rightarrow$ Many

Machine Translation
(English $\rightarrow$ Dutch)



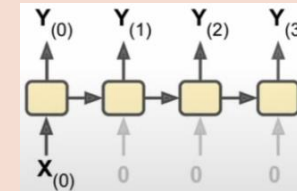
Many $\rightarrow$ One

Sentiment Analysis
(text $\rightarrow$ positive/negative)



One $\rightarrow$ Many

Image Captioning
(image $\rightarrow$ description)



Application Examples

Time Series

Weather forecasting
Stock price prediction

Text

Sentiment analysis
Text generation
Translation

Speech

Speech to text
Voice assistants

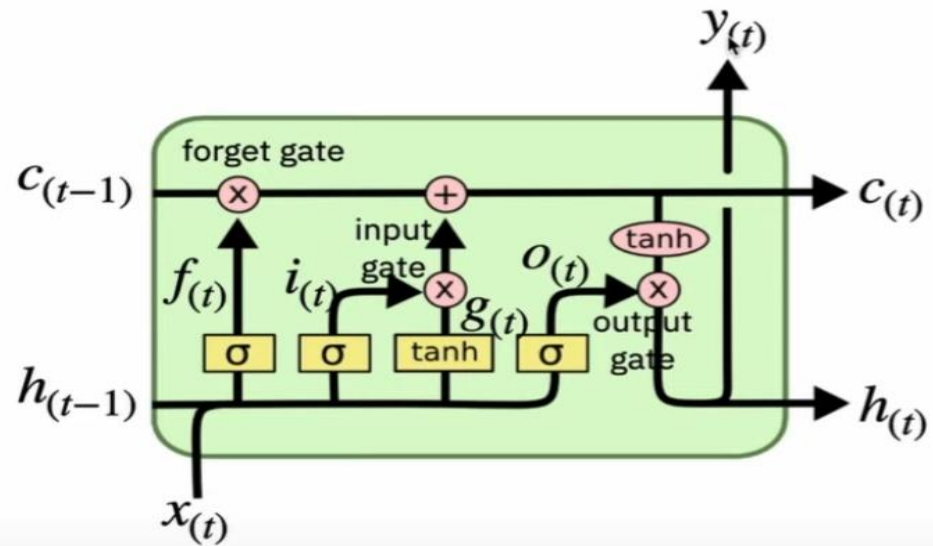
Video

Action recognition
Video description

Limitations & Improved RNNs

Vanishing Gradient Problem

In long sequences, gradients shrink exponentially during backpropagation, causing the network to **forget** early inputs and fail to learn **long range dependencies**.



LSTM: Long Short-Term Memory

- **Memory cells:** store long term information
- **Forget gate:** what to erase from cell state
- **Input gate:** what new info to write
- **Output gate:** what to expose as h_t

GRU (Gated Recurrent Unit): simplified LSTM with fewer parameters. Often trains faster than LSTM.

Bidirectional RNNs: process sequences both forward and backward, giving access to both past and future context.